

KARTHIK MANNEM

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TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Gradient Boosting, Time-Series Forecasting, Backtesting
NLP & LLM: Transformers, HuggingFace, LangChain, RAG, Tool-Calling Agents, Prompt Engineering, Eval Harnesses
Data: Pandas, NumPy, Feature Engineering, PostgreSQL, MySQL, MongoDB, ChromaDB / Vector Search
MLOps & Cloud: AWS (SageMaker, Bedrock, Lambda, API Gateway, S3), Terraform, Docker, GitHub Actions, Flask

EXPERIENCE

- AI Model Evaluation Analyst

Remote

Handshake AI

Sep 2025 – Apr 2026

 - Scored 1,000+ LLM responses against rubrics for factual accuracy, reasoning quality, and instruction-following.
 - Catalogued recurring failure modes and hallucination patterns into reusable categories that fed prompt-refinement workflows and tightened rubrics where graders disagreed.
 - Sustained high inter-annotator agreement and escalated ambiguous rubric cases, improving benchmark reliability.
- Machine Learning Engineer (Freelance)

Remote

Self-Employed

Aug 2023 – Jun 2024

 - Delivered deep learning models (LSTM, CNN) for 3 client applications, owning preprocessing, feature engineering, cross-validated training, and the deployed inference endpoint.
 - Containerized ML services with Docker behind Flask REST APIs for reproducible deployment and real-time prediction.
 - Cut training time by roughly 25% through pipeline profiling, batched preprocessing, and training-config tuning.

PROJECTS

- Retail Demand Intelligence Platform – Forecasting, Anomaly Detection & Agentic Q&A

GitHub

Python, scikit-learn, Flask, Anthropic/OpenAI/Bedrock, Terraform, AWS, Docker, GitHub Actions

Live Demo

 - Built a 3-model pipeline (seasonal-naive, Holt-Winters, GBM) over 240 series, rolling-origin backtested, selecting a winner per series (GBM: 7.9% WAPE, best on 154/240).
 - Shipped 80% prediction intervals and occlusion-based feature attribution, reporting measured coverage on held-out data (74.2% vs. 80% nominal) instead of asserting calibration.
 - Designed a 6-tool agentic Q&A layer over a pluggable LLM backend with audit logging, retry/backoff, token-cost tracking, and a 37-case eval harness scoring tool routing in CI.
 - Closed the MLOps loop: drift detection triggers per-series re-selection with a logged audit trail; deployed to AWS (SageMaker, Lambda, Bedrock) via Terraform, Docker, and CI.
- Preventive Care Recommendation Agent – Agentic RAG for Clinical Decision Support

GitHub

Python, LangChain, ChromaDB, Llama-3.1, Streamlit

 - Built a LangChain RAG pipeline over a ChromaDB index of cardiovascular and metabolic guideline PDFs, so recommendations cite retrieved text rather than model recall.
 - Combined a computed Framingham risk score with retrieved guidelines into a structured, risk-specific screening and vaccination plan.

OPEN-SOURCE CONTRIBUTIONS

unionai-oss/pandera – 1 PR merged: #2437 supports itemsize-parameterized NumPy dtypes (S16, U16) in the pandas engine, with tests.
pytorch/pytorch – 1 PR approved, open for merge: #191687 fixes torch.fx codegen for keyword/escaped module names, with tests.

EDUCATION

- California State University, Northridge

Los Angeles, CA

Master of Science in Computer Science | CGPA: 3.92 / 4.0

Aug 2024 – May 2026
- Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Chennai, India

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.21 / 10.00

Aug 2020 – May 2024

LEADERSHIP & CERTIFICATIONS

Board of Directors, Facilities & Ops – USU, CSUN (2024–26): 38,000+ students, \$5M budget.
Certifications: AWS AI & ML (Foundation, Associate) · NVIDIA Deep Learning Fundamentals